

Full Stack Development with MERN

Project Documentation

1. Introduction

- **Project Title:** Online Complaint Registration Portal
- **TeamMembers:** PolumuruGanesh (**INDIVIDUAL PROJECT**)

2. Project Overview

- **Purpose:** To streamline the process of lodging and managing citizen complaints, ensuring transparency and quick resolution across various government departments.
- **Features:**
 - Role-based dashboards (Citizen, Officer, Admin).
 - Secure Auth for Citizen sign-up and login.
 - Capability to lodge grievances directly to 15 different Government Departments.
 - Real-time status tracking with badges (Review, Assign, Resolve).
 - Officer dashboard to view 20 active complaints with a click-to-view modal mechanism.
 - Global oversight functionality for Admins.

3. Architecture

- **Frontend:** Built using React.js, HTML5, CSS3, and JavaScript (ES6+) to create a responsive user interface.
- **Backend:** Developed with Node.js and Express.js to handle robust server-side logic and RESTful APIs.
- **Database:** MongoDB used alongside Mongoose ODM for flexible schema design and database interactions.

4. Setup Instructions

- **Prerequisites:** Node.js (v18+ or v20+), MongoDB Compass or Local MongoDB Community Server, Git, and WebStorm IDE.
- **Installation:**
 1. Clone the project repository using Git.
 2. Navigate to the backend/ directory, run npm install, and configure config/db.js for database connection.
 3. Navigate to the frontend/ directory and run npm install to install React dependencies.

5. Folder Structure

- **Client:** The frontend/ directory contains public/index.html (Base HTML) and src/ which holds pages/ (View Dashboards for User, Admin, Officer), App.jsx (Core Routing), index.css (Global Styling), and main.jsx (DOM Bootstrapper).
- **Server:** The backend/ directory contains config/db.js (MongoDB Config), models/ (User & Complaint Schemas), routes/ (Modular Express API Routes), and server.js (Main Backend Entry Point).

6. Running the Application

- **Frontend:** npm run dev (or npm start) in the client (frontend) directory.

- **Backend:** npm start in the server (backend) directory.

7. API Documentation

- **Endpoints:**
 - POST /api/auth/signup - Register a new Citizen/Officer.
 - POST /api/auth/login - Authenticate and login users.
 - POST /api/complaints - Submit a new grievance.
 - GET /api/complaints - Fetch complaints based on role access (Citizen: latest 10, Officer/Admin: up to 20 active).

8. Authentication

- Implements Secure Auth for all users (Citizens, Officers, Admins) using full credentials. Standard JWT (JSON Web Tokens) can be utilized to maintain secure sessions and protect dashboard routes.

10. Testing

- Manual UI and functional testing. API endpoints are tested and verified using tools like Postman.

11. Screenshots or Demo

- **GITHUB REPOSITORY:** -<https://github.com/Ganesh-raj998/Online-complaint-registration->
- **Demo video link**
:-<https://drive.google.com/file/d/1Id2GX5ttvTwrZJ89duFvYBj9CeF8mJJ/view?usp=drivesdk>

• 12. Known Issues

- None documented currently.

13. Future Enhancements

- Implementing automated Email/SMS notifications for real-time status updates.
- Integrating AI-based automated routing to assign complaints to the exact sub-department faster.